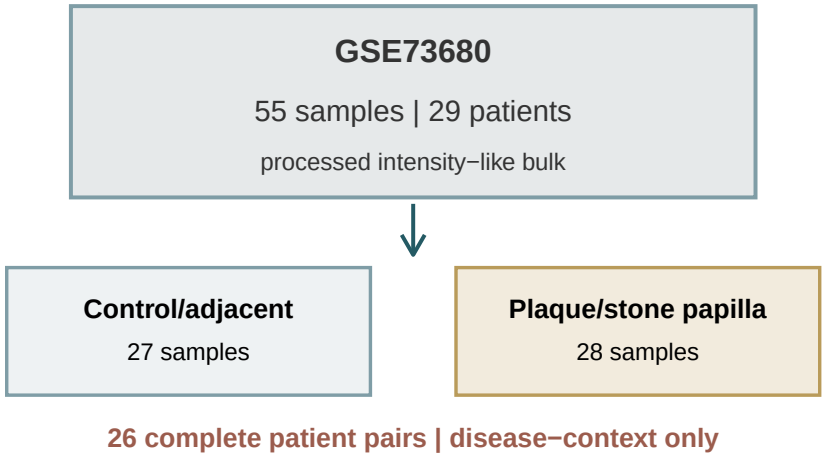


Bulk papillary disease context and tissue-state sensitivity

GSE73680 supports paired module-level disease-context consistency; causal genes, plaque mechanisms and cell fractions are not inferred

A

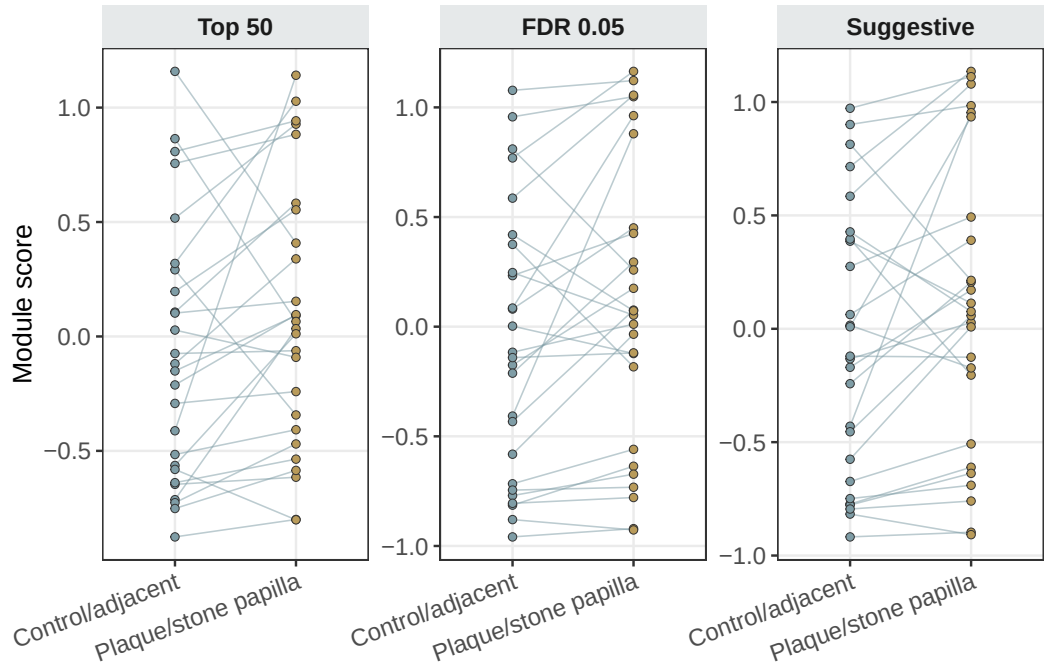
Paired bulk design



B

Paired MAGMA module scores

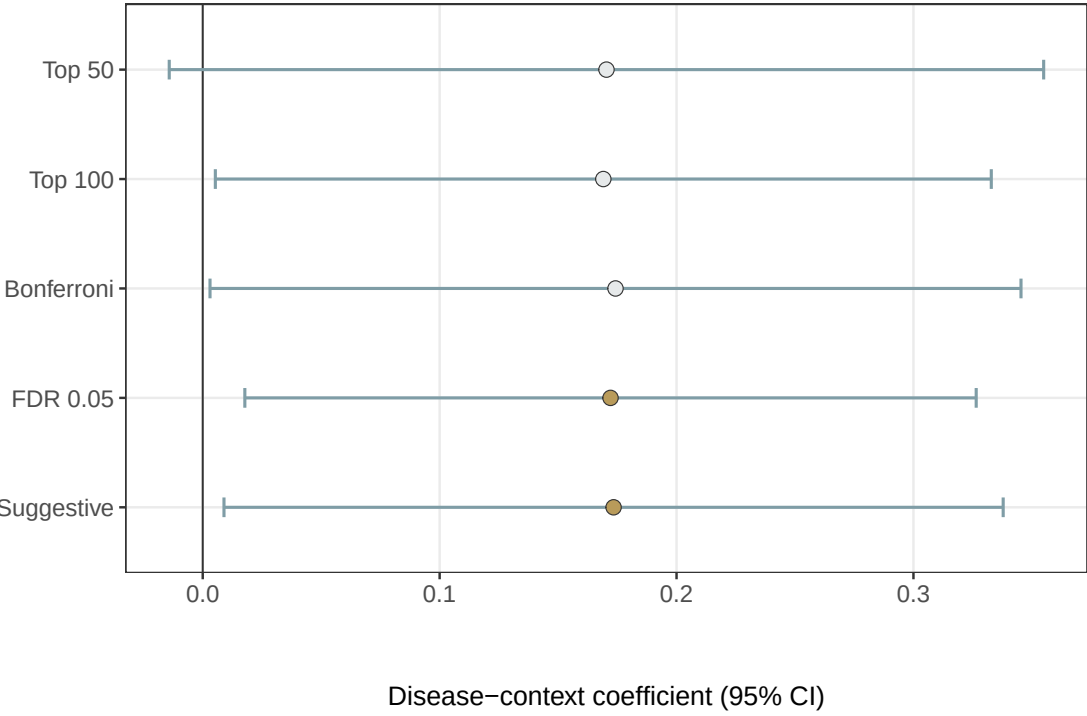
Selected canonical modules; 26 patient pairs



C

Base paired group coefficients

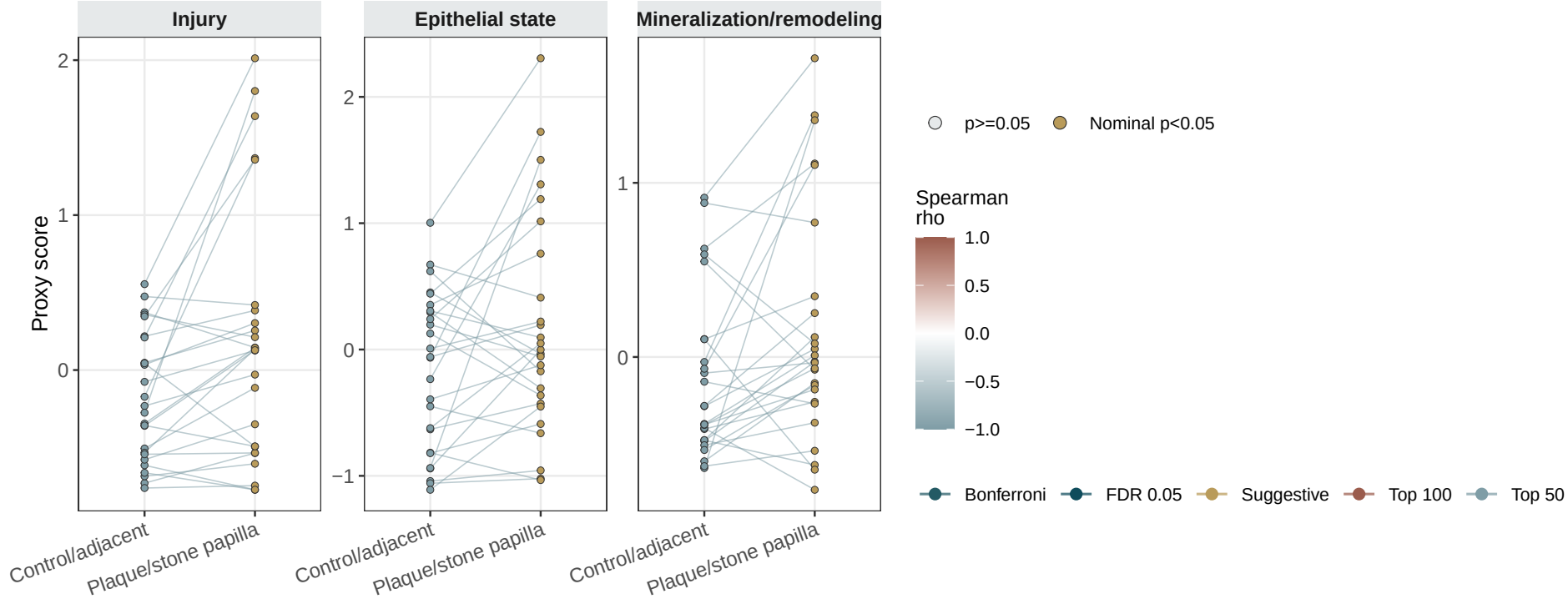
Patient fixed effects; all module FDR values >0.05



D

Tissue-state proxy shifts

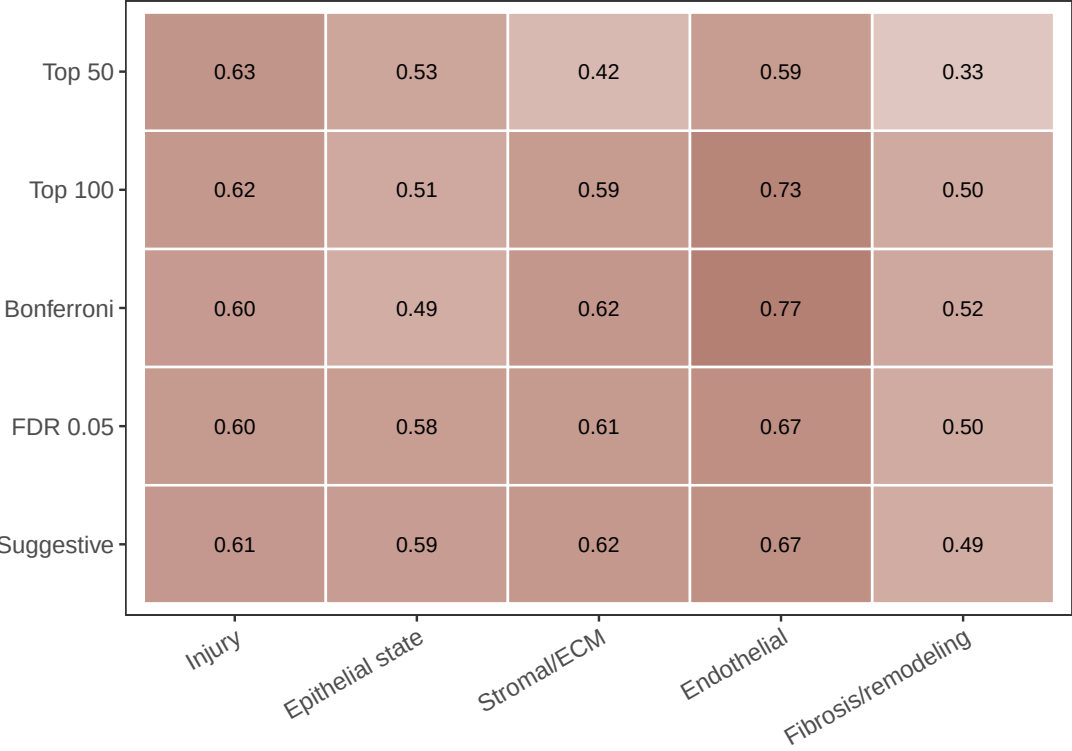
Selected marker scores; proxies are not cell fractions



E

Module-tissue-state coupling

Paired disease-minus-control differences; non-causal



F

Tissue-state sensitivity

Attenuation indicates disease-state embedding

